

Flavio Romano

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Software Engineer and Technical Lead specializing in backend systems, IAM solutions, and scalable cloud-native architectures. Experienced in leading cross-functional teams, designing distributed systems, and delivering enterprise-grade software in telecommunications and sustainability domains.

Currently pursuing an MSc in Artificial Intelligence while working part-time at NTT DATA, strengthening a software engineering background with advanced AI and machine learning expertise. Passionate about applying AI to solve complex industrial problems at scale.

Experience

Software Engineer, NTT DATA – Pisa, IT **March 2024 – Present**

Identity & Access Management Platform (Telco Client) Aug 2024 – May 2025

- Contributed to the design and implementation of OAuth 2.0, OpenID Connect, and SAML-based authentication solutions for a leading Italian telecommunications provider.
- Enhanced a customized Keycloak deployment, enabling adoption of the Phantom Token architecture to improve security, scalability, and operational flexibility.
- Developed Spring Boot microservices supporting authorization workflows, including token introspection, redirect management, and third-party JWT/JWE processing.
- Collaborated within a 5-person engineering team delivering authentication and identity management capabilities across multiple enterprise portals.

Logistics Data Processing Platform May 2025 – Sep 2025

- Designed and implemented Spring Batch jobs for large-scale shipment processing, handling hundreds of thousands of records through complex transformation and enrichment pipelines.
- Developed resilient batch-processing workflows with a focus on reliability, maintainability, and performance optimization.

AI-powered Due Diligence Platform for Sustainability Compliance Sep 2025 – Jan 2026

- Served as Technical Lead for the development of an AI-powered platform designed to simplify the generation of due diligence documentation required by European import/export sustainability regulations.
- Coordinated a team of 4 engineers, managing task allocation, technical mentoring, and knowledge-transfer sessions.
- Designed and implemented backend services using Spring Boot, Redis caching, and optimized offset- and cursor-based pagination strategies to support large datasets efficiently.
- Worked closely with stakeholders to translate regulatory and business requirements into scalable software solutions.

Corporate Sustainability & CO₂ Emissions Assessment Platform Jan 2026 – Present

- Acting as Technical Lead for the development of a platform that estimates CO₂ emissions generated by software systems, hardware infrastructure, and AI workloads.
- Leading a team of 5 engineers, coordinating daily stand-ups, milestone planning, task assignment, code reviews, and technical decision-making.
- Serving as the primary technical point of contact for the client, gathering requirements and aligning technical deliverables with business objectives.
- Designing scalable backend services and data-processing workflows to support sustainability reporting and environmental impact assessment.

Software Engineer Stage, NTT DATA – Pisa, IT **Dec 2023 – March 2024**

- Developed a Java microservice for the GreenIT ecosystem to collect hardware metrics and estimate Software Carbon Intensity (SCI) and CO₂ emissions on Google Cloud Platform infrastructures.

Technical Skills

Programming languages: Bash, C, Golang, Java, JS/TS, Python, SQL

Framework: Gin, Pandas, Selenium, Spring, SvelteKit, Pytorch, Keras

Cloud provider: AWS, GCP

Projects

Encoding Classification using LSTM (github)

Apr 2026 – May 2026

- Developed a PyTorch-based sequence classification model to automatically identify the encoding algorithm applied to a text string, distinguishing between Caesar Cipher, ROT13, and MD5 transformations.
- Built and trained an LSTM neural network with approximately 230k trainable parameters using a dataset of more than 72,000 samples derived from English-language anagrams.
- Designed an NLP preprocessing pipeline based on token embeddings and sequence padding, achieving 97.99
- Implemented automated training and inference workflows, enabling real-time prediction of encoding types from arbitrary input strings.

Multi-output Regression for Noisy Datasets

Oct 2025 – Jan 2026

- Developed a deep learning pipeline for a multi-output regression task using a highly noisy dataset. Achieved significant error reduction on the test set through rigorous data preprocessing and feature engineering.
- Conducted systematic model selection and hyperparameter tuning using Optuna, comparing various Neural Network architectures implemented with Keras and Scikit-learn.
- Performed a robust final model assessment to ensure generalization, earning high marks for the methodological approach in the Machine Learning graduate course.

Coinquilining (github)

Aug 2023 - Dec 2023

- Developed from scratch as final project for the "Web App Development" course, earning a top grade (30/30). Coinquilining is a PWA for managing roommate life, featuring shared fridge and pantry management, expense tracking, and recipe suggestions.
- Implemented expiration tracking with notifications, ingredient-based recipe searches using Puppeteer to scrape a dynamic site, and a calendar for fair task distribution (time shift problem).
- Tech Stack: Supabase, TypeScript, SvelteKit, Tailwind CSS, Puppeteer

Education

University of Pisa, Master of Science (MSc) in Artificial Intelligence

Sept 2025 – Present

University of Pisa, Bachelor of Computer Science

Sept 2020 – May 2024

- **Thesis:** Development of a configurable connector to monitoring SCI and CO2 emissions of GreenIT® Google Cloud Platform components.